Lesson notes · for you

Pass it on: mesh networks

IoT · Level 4 · Lesson 2

Materials
Paper, a pencil and a ruler; A piece of string and scissors; Optional: friends, to try a human mesh outdoors

Learning goals

- Explain how a mesh passes messages hop by hop to a gateway.
- Find the weak points in a network and fix them with relays.
- Say why a mesh keeps working when one device fails.

Activity: Paper mesh

1. Draw a place you know, such as your school or a farm. Mark ten sensors as dots and the gateway as a square. Choose a scale, such as 1 cm for 20 m.
2. At that scale, a 200 m radio range is 10 cm of string. Cut it carefully, away from other people.
3. Hold one end on each sensor in turn and draw a line to every point the other end reaches. Each line is a radio link.
4. Circle any sensor with no line (cut off) and any with only one line (one fallen neighbour would cut it off).
5. Add the fewest relays that give every sensor a route, and two routes where you can. Write each sensor's hop count beside it.

Use scissors carefully. If you try a human mesh outside, stay in a safe open space away from roads, and pass notes only to someone within arm's reach.

Discussion

- What happened to your network when you removed one sensor?
- Why do sensors near the gateway run down first?
- Where in your town would a mesh help, and what could break it?

Check yourself

- Why does a mesh keep working when one sensor fails?

Try it this week
Paper mesh: draw ten sensors and a gateway on a map, link every pair that a string as long as the radio's range can reach, find a sensor that's cut off, and add a relay to rescue it.

Worksheet

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Name _____ Class _____ Date _____

1 **Zainab's sensor by the river has only one neighbour, and floods keep knocking that neighbour over. What is the best fix?**

- ◯ Add a relay on higher ground
- ◯ Give her sensor a bigger battery
- ◯ Make her sensor send more often

2 **In a mesh, messages can route round a broken sensor, but if the gateway fails, no message gets out at all.**

- ◯ True
- ◯ False

3 **Match each weak point to its fix.**

A gap wider than the radio's range	• • • •	Give it a solar panel and a battery
A sensor with only one neighbour	• • • •	Put a relay in the gap
The gateway loses power at night	• • • •	Give the busiest nodes bigger batteries
Sensors next to the gateway run flat first	• • • •	Add a relay as a second neighbour

4 **In a farm mesh, every sensor sends one reading an hour. Predict: whose batteries run down first?**

- ◯ The sensors next to the gateway
- ◯ All of them at the same time
- ◯ The sensors farthest from the gateway

5 **Count the hops along a line of sensors.**

Sensors stand every 250 m along a road, and each radio reaches 300 m. A message from a sensor 1,000 m from the gateway needs _____ hops. If each hop adds 0.1 s, it arrives after about _____ s.

Word bank: 3 · 1 · 4 · 0.4

✎ **Where near you could a mesh help: a farm, a river, a market, a school compound? Where would you put the gateway, and what might knock a sensor over?**

Check your answers

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1 **Add a relay on higher ground**

Her sensor's problem is its single link, not its power.

2 **True**

Every route ends at the gateway, so it's the one part a mesh can't route round.

3 **A gap wider than the radio's range — Put a relay in the gap**
A sensor with only one neighbour — Add a relay as a second neighbour
The gateway loses power at night — Give it a solar panel and a battery
Sensors next to the gateway run flat first — Give the busiest nodes bigger batteries

Each fix targets its cause: relays add the links a gap or a lonely sensor is missing, the gateway needs power that lasts the night, and the busiest relays need the most battery.

4 **The sensors next to the gateway**

Every relay spends battery, and the sensors next to the gateway pass on everyone's messages as well as their own.

5 **Sensors stand every 250 m along a road, and each radio reaches 300 m. A message from a sensor 1,000 m from the gateway needs 4 hops. If each hop adds 0.1 s, it arrives after about 0.4 s.**

With 300 m of range, each sensor reaches only the next one, 250 m away, so the message makes 1,000 ÷ 250 = 4 hops, and 4 × 0.1 = 0.4 s.